

Class: EventController

Responsibilities:

- Fetch all events and filtered events
- Fetch organizer-specific events
- Create, update, and delete events
- Map Firestore documents to Event models
- Provide real-time event update listeners

Collaborators:

- FirebaseFirestore
- Event
- AppSession
- QuerySnapshot
- ListenerRegistration

Class: WaitingListController

Responsibilities:

- Join and leave waiting lists
- Run lottery and mark entrants as selected
- Redraw next entrant when a selected entrant declines
- Update entrant statuses (pending, selected, accepted, cancelled)
- Fetch waiting list entries and counts
- Promote next entrant automatically
- Emit notifications during status transitions

Collaborators:

- FirebaseFirestore
- WaitingListEntry
- Event
- NotificationController
- WriteBatch
- DocumentSnapshot
- QuerySnapshot

Class: NotificationController

Responsibilities:

- Create offer and generic notifications
- Save notifications to Firestore
- Delete notifications
- Fetch notifications for entrants, organizers, and admins
- Map Firestore snapshots into Notification models

Collaborators:

- FirebaseFirestore
- Notification

- User
- Event
- UserController
- DocumentReference
- QuerySnapshot

Class: UserController

Responsibilities:

- Save and update user profiles
- Fetch user profiles by ID
- Delete users
- Fetch all users by role
- Map Firestore snapshots to User (Entrant/Organizer/Admin)
- Provide entrant history (cross-collection)

Collaborators:

- FirebaseFirestore
- User
- Entrant
- Organizer
- Admin
- WaitingListEntry
- EntrantHistoryItem
- DocumentSnapshot
- QuerySnapshot

Class: ImageController

Responsibilities:

- Fetch all images or a single image
- Create/upload image documents
- Delete stored images
- Convert snapshots into Image models

Collaborators:

- FirebaseFirestore
- Image
- QuerySnapshot

Class: Event

Responsibilities:

- Represent event data (name, description, dates, tags, capacity)
- Serialize to/from Firestore (toMap, fromSnapshot)
- Store lottery_done state
- Provide QR code payload for the event

Collaborators:

- DocumentSnapshot
- Date
- List<String>

Class: WaitingListEntry

Responsibilities:

- Represent an entrant's waiting list record
- Store status, timestamp, eventId, entrantId, eventName
- Deserialize from Firestore safely

Collaborators:

- DocumentSnapshot
- DocumentReference
- Date

Class: Notification

Responsibilities:

- Represent an in-app notification
- Store sender, recipients, message, timestamp, related event
- Provide display helpers (eventName, createdAt)

Collaborators:

- User
- Event
- ArrayList<User>
- Date

Class: User (abstract)

Responsibilities:

- Base type for Entrant/Organizer/Admin
- Store id, name, email, phone, notificationsEnabled
- Enforce getRole() for subclasses

Collaborators:

- None directly (subclassed by Entrant/Organizer/Admin)

Class: Entrant

Responsibilities:

- Represent entrant user and extend User behavior

Collaborators:

- User

Class: Organizer

Responsibilities:

- Represent organizer user and extend User behavior

Collaborators:

- User

Class: Admin

Responsibilities:

- Represent admin user and extend User behavior

Collaborators:

- User

Class: Image

Responsibilities:

- Represent image metadata and base64 content
- Decode base64 to Bitmap
- Map Firestore snapshot → Image model

Collaborators:

- DocumentSnapshot
- BitmapFactory

Class: EntrantHistoryItem

Responsibilities:

- Represent a read-only history entry for an entrant
- Store eventId, eventName, status, timestamp

Collaborators:

- Date

Class: BaseDashboardActivity

Responsibilities:

- Provide unified dashboard structure (toolbar, bottom nav, fragment swapping)
- Handle logout logic and navigation to login
- Manage toolbar title and back button behaviour
- Delegate tab → fragment selection to subclasses

Collaborators:

- AppSession
- LoginActivity
- Fragment (subclasses define)
- BottomNavigationView

Class: WaitingListActivity

Responsibilities:

- Display all entrant statuses for a specific event
- Subscribe to Firestore for pending/selected/accepted/cancelled groups
- Run organizer lottery and handle redraw logic
- Maintain counts and UI visibility for statuses

Collaborators:

- WaitingListController
- FirebaseFirestore
- SimpleEntrantAdapter
- EntrantRow

Class: SplashActivity

Responsibilities:

- Auto-login user via deviceId
- Route user to correct dashboard based on role
- Show login screen if session missing or invalid

Collaborators:

- UserController
- FirebaseFirestore
- AdminActivity / OrganizerActivity / EntrantActivity
- LoginActivity
- AppSession

Class: SignupActivity

Responsibilities:

- Create new entrant accounts
- Validate inputs and check email uniqueness
- Save user to Firestore
- Route user to login after signup

Collaborators:

- FirebaseFirestore
- LoginActivity
- AppSession

Class: SimpleEntrantAdapter

Responsibilities:

- Render a simple entrant row (name, status, joined date)
- Refresh dataset when new Firestore snapshot arrives

Collaborators:

- EntrantRow
- RecyclerView

Class: NotificationAdapter

Responsibilities:

- Render rich notification cards for entrants/organizers
- Format timestamp and recipient list
- Notify listener on click

Collaborators:

- Notification
- User
- Event
- OnNotificationClickListener

Class: EntrantNotificationAdapter

Responsibilities:

- Display simple entrant notifications (event, message, date)
- Bind basic UI fields for notifications

Collaborators:

- Notification
- RecyclerView

Class: AdminNotificationAdapter

Responsibilities:

- Render admin-facing notification rows
- Provide delete button and invoke deletion callback
- Format sender, recipients, event name, timestamp

Collaborators:

- Notification
- User
- DeleteListener
- RecyclerView

Class: AdminPagerAdapter

Responsibilities:

- Supply admin fragments (Events / Images / Profiles) to ViewPager
- Instantiate correct fragment for each tab

Collaborators:

- AdminEventListFragment
- AdminImageListFragment

- AdminProfileListFragment

Class: OrganizerPagerAdapter

Responsibilities:

- Provide organizer tabs (My Events / Create Event)
- Map each tab → appropriate fragment

Collaborators:

- OrganizerEventListFragment
- OrganizerCreateEventFragment

Class: AppSession

Responsibilities:

- Store the currently authenticated user in memory
- Expose cached userId, role, and email
- Preserve launching intent for process-death recovery
- Provide singleton Firestore instance
- Initialize/clear global session state

Collaborators:

- User
- FirebaseUtils
- FirebaseFirestore
- Intent

Class: FirebaseUtils

Responsibilities:

- Provide helper wrappers around Firestore CRUD operations
- Create/set/merge documents
- Query documents with whereEqualTo
- Delete documents and batches
- Check document existence
- Retrieve typed objects from Firestore

Collaborators:

- FirebaseFirestore
- DocumentSnapshot
- WriteBatch
- onSuccessListener / onFailureListener
- QuerySnapshot

Class: ImageUtil

Responsibilities:

- Convert Bitmap → Base64 and Base64 → Bitmap

- Extract filename from URI
- Utility helpers for image handling in Android

Collaborators:

- Bitmap
- Uri
- Base64
- ContentResolver

Class: QrUtil

Responsibilities:

- Generate QR code bitmaps from payload strings
- Save generated QR code/image bitmaps to phone gallery

Collaborators:

- Bitmap
- MultiFormatWriter
- MediaStore
- ContentResolver

Class: DateUtil

Responsibilities:

- Format event date ranges for UI display
- Format individual Date objects into readable strings

Collaborators:

- Event
- Date
- SimpleDateFormat

Class: DeletionConfirmationUtil

Responsibilities:

- Build and show reusable deletion confirmation dialogs
- Execute callback when user confirms deletion

Collaborators:

- AlertDialog
- Context